

Task 1: Order Flow Imbalances

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Question 1. What's the motivation behind measuring OFI at multiple depth levels of the order book?

The motivation for measuring *Order Flow Imbalance (OFI)* at multiple depth levels of the order book is to capture richer and more nuanced information about supply and demand pressure that cannot be fully observed at just the Level 1 data. The key reasons are:

1. **Market pressure exists beyond the best quotes:** Level-1 OFI only reflects surface liquidity. Traders often place, cancel, or modify orders at deeper levels (Level 2 to 10) to conceal trading intentions or manage risk. Thus, deeper-level OFIs can reveal hidden buying/selling pressure that may lead to future price moves.
2. **Improved predictive power:** Empirical evidence shows that including deeper-level OFIs significantly improves the ability to forecast short-term price changes. These levels often react earlier to large trades and evolving market sentiment, offering valuable leading indicators.

Relying solely on Level-1 data risks discarding informative microstructural signals, especially in high-frequency or low-liquidity settings. Deeper levels provide a more complete picture of order book dynamics.
3. **Enabling robust signal integration:** With multi-level OFIs, dimensionality reduction methods like *Principal Component Analysis (PCA)* can be applied to extract dominant patterns of imbalance. The resulting integrated OFI is often more stable and informative than any single-level OFI.

Question 2. Why do the authors use Lasso regression rather than OLS for estimating cross-impact?

The authors use Lasso regression rather than OLS for estimating cross-impact because Lasso addresses two key problems inherent in high-dimensional cross-asset modeling:

1. **High dimensionality ($N > T$):** When the number of assets N is large and the time window for estimation is short (e.g., a 30-minute window of 1-minute returns), the number of predictors exceeds the number of

observations. In such underdetermined systems, OLS becomes ill-posed, leading to unstable and non-unique solutions.

2. **Multicollinearity among OFI features:** A significant portion of cross-asset OFIs are highly correlated, violating OLS assumptions and inflating the variance of estimated coefficients. The paper shows that over 10 percent of OFI correlations exceed 0.3, which introduces strong multicollinearity.
3. Lasso regression resolves both issues by imposing L1 regularization, which performs variable selection and coefficient shrinkage and producing sparse and stable estimates of cross-impact coefficients.

Question 3. Why is OFI considered a better predictor of short-term returns than trade volume?

OFI is widely regarded as a better predictor of short-term returns compared to trade volume due to the following reasons:

1. **OFI captures order book dynamics beyond executed trades:** While trade volume only accounts for completed transactions, OFI reflects the net effect of order submissions, cancellations, and modifications at the bid and ask, offering a richer view of supply and demand pressure.
2. **OFI includes information from passive liquidity:** Changes in limit orders and cancellations signal evolving market sentiment. OFI quantifies these changes, whereas trade volume does not.
3. **OFI measures directionality, not just activity:** Trade volume is inherently symmetric—every trade has a buyer and a seller—so it lacks directional insight. OFI, on the other hand, reflects the imbalance between buying and selling interest, which more directly drives price changes.